

- * Completed the first major maintenance window 12-15 hour day
- * Fixed network configuration consistency across an AI fleet.
- * Standardized hostnames by GPU type and BMC/XCC access.
- * Standardized DHCP/netplan behavior
- * Isolated rack focused for maintenance from the power spaghetti
- * Re-cabled and cleaned the entire rack.
- * Improved airflow.
- * Reduced operational risk.
- * Verified inventory
- * Created actual documentation evidence with before/after photos.
- * Kicked off multi-terabyte migrations.
- * Solved the BTRFS mounting problem.
- * Evaluated workstation strategy.
- * Evaluated storage architecture strategy.
- * Got management visibility
- * Got maintenance authority enforced

// fixing the inherited reality

- * accumulated expediency
- * everybody plugging things in
- * no lifecycle management ownership
- * airflow impudence as an afterthought

// Identified:

- * rack too shallow
- * PDU placement wrong
- * power constraints
- * mixed ownership
- * missing blanks
- * missing cable management accessories

// Scope of work

- > Capacity planning.
- > Resource allocating.
- > Hardware lifecycle management.
- > Migration strategy.
- > Disaster recovery.

// Team communication post maintenance window

6/5 8:52 PM

Team,

This first maintenance is now complete, it's been a long day and extended configurations for everything.

Please update your records for the systems:

hostname changes to reflect gpu for faster identification
ip changes due to dhcp to specified ports

[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]

network configuration for os and for system bmc/xcc has been modified to be consistent across the fleet of systems

backend cabling of [REDACTED]-h200, [REDACTED]-h200, [REDACTED]-mi300x (all rack 1) have been recabled to remedy the thermal/airflow issue, in addition to detached from the others from an intertwined power stance

PLEASE DO NOT CONNECT TO:

[REDACTED]-h200 / [REDACTED]
[REDACTED]-h200 / [REDACTED]
as they are still doing data migration efforts

[REDACTED]-b200 / [REDACTED] has finished it's data migration

Once the data migration is finished/verified for :

[REDACTED]-h200 / [REDACTED]
[REDACTED]-h200 / [REDACTED]
[REDACTED]-b200 / [REDACTED]

I still need to reconfigure the systems to prepare for the requested drives removal.

The other systems should be available.
Please log into those systems to verify and let me know if you have concerns.

I will send a follow-up update to when

[REDACTED]-h200 / [REDACTED]
[REDACTED]-h200 / [REDACTED]
[REDACTED]-b200 / [REDACTED]
are available for use again

===

6/8 8:06 AM

Good Morning team,

As mentioned on Friday evening (US est),
PLEASE DO NOT CONNECT TO:

██████-h200 / ██████
██████-h200 / ██████
as they are still doing data migration efforts

██████-b200 / ██████ has finished it's data migration

===

Data migration efforts are in progress, if you connected, please stop the workload and log off, so the data migration efforts can continue, else whatever data migration efforts going on will be incomplete/corrupted.

It seems

██████-h200 / ██████
██████-h200 / ██████
have workloads active.

These migrations needs to be complete, so that the respective users can verify/validate their data before Wednesday 6/10 eod (US est) so that I can prepare the system reconfiguration to clean up the former drive shares to remove them and return them

If you are using

██████-b200 / ██████
Please login to verify your data, in the new data-migration share and let me know if you have issues before Wednesday 6/10.

6/10 10:31 AM

Team,

The data-migration efforts should be done. Users of the affected systems (██████) have been reached out to have them verify/validate the data to address any issues before this Thursday 6/11/26 @4pm (US est) as the 4x 30tb drives will be wiped, system reconfigured, and drives pulled to return to the equipment owner on Friday 6/12.

Reminder for users of ██████) and ██████
notification has been sent to complete any workloads by Friday 6/12/2026 @ 4pm (US est), the system will be powered down as the equipment owner has requested to have the pcie network adapters returned. Request has been deferred to early next week.

6/12 3:29 PM

Team,

Systems (██████-h200, ██████-h200, ██████-b200) had the 4x 30tb drives wiped, system reconfigured, and drives pulled and returned to the equipment owner through proxy (handed off to ██████)

██████-h200 and ██████-h200 are back online and available for use.

I will be shutting down systems (██████-b200 and ██████-mi300x) on Monday 6/15/2026 morning by 9am US est at latest, and let the equipment owner know they can come pull those pcie network adapters. I'll send out a follow-up when they are done and I've powered on the systems.

6/15 1:05 PM
Team,

Systems (██████-b200 and ██████-mi300x) are back online and available. Equipment owner has reclaimed the pcie network adapters.

6/22 11:52 AM
Team,

I've been informed that you have a testing cycle coming up, and it's recommended to do a short notice maintenance, planned for this Friday 6/26/2026 @ 12 noon US est. The work will be to decouple the power connections from the 5 systems :

██████-h200 / ██████
██████-h100 / ██████
██████-b200 / ██████
██████-b200 / ██████
██████-mi300x / ██████

This is to ensure that future maintenance on any one of those systems is just isolated to that specific system and not affecting the other intertwined systems.

Please voice any issues, concerns, questions, objections before end of day 6/24/2026 Wednesday @4pm US est. Else maintenance will proceed as scheduled, the systems will be offline until maintenance is complete and communicated as such.

Please ensure all workloads are completed and do not start any new workloads after 6/25/2026 Thursday @ 4pm.
Thanks

kt

6/22 4:00 PM

Team,

██████-mi300x / ██████ : storage reconfiguration
This system is getting its storage drives reconfigured
Users of this system have also been notified via email to

BACK UP AND VALIDATE YOUR DATA

Please do so by Friday 6/26/2026 by 4pm US est, after this the system storage capacity will be reconfigured for data2, all data destroyed and drives pulled and repurposed in another system.

thanks,

kt

6/26 12:02 PM

Team,

Maintenance is starting to decouple the power connections from the 5 systems :

██████-h200 / ██████
██████-h100 / ██████
██████-b200 / ██████
██████-b200 / ██████
██████-mi300x / ██████

This is to ensure that future maintenance on any one of those systems is just isolated to that specific system and not affecting the other intertwined systems.

██████-mi300x / ██████ : storage reconfiguration
This system is getting its storage drives reconfigured and excess drives repurposed to ██████-h100 / ██████

Affected systems will be down until maintenance completion is communicated again.

6/26 5:43 PM

Team,

The maintenance to decouple the power connections from the 5 systems :

██████-h200 / ██████
██████-h100 / ██████
██████-b200 / ██████
██████-b200 / ██████
██████-mi300x / ██████

is complete.

//note the ip change for the following systems

██████-h100 > ██████████

Storage reconfiguration of ██████████-mi300x / ██████████ is complete.
Storage drives reconfigured repurposed to ██████████-h100 / ██████████

Systems should be back up, please login to verify your data.

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```
//maintenance started as scheduled with no objections raised during window of opportunity
//maintenance executing as scheduled for 6/26/2026 at 12pm US est
//users logged in with active workloads
```

```
IPv4 address for ens11f0np0: [REDACTED]
IPv6 address for ens11f0np0: fd55:faaf:e1ab:2021:ba3f:d2ff:fe90:addc

=> / is using 91.1% of 195.80GB
```

```
[REDACTED]-h200:~$
16:14:42 up 20 days, 19:27, 10 users, load average: 20.68, 20.82, 21.63
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
[REDACTED] pts/21    tmux(47088).%6  10Jun26  7:34m   0.08s   0.03s -bash
[REDACTED] pts/24    tmux(47088).%0  08Jun26 17days  1.42s   1.42s -bash
[REDACTED] pts/16    tmux(47088).%1  10Jun26  2:24m  58.28s  33.50s /data2/
[REDACTED] /miniconda3/envs/gemma4_qkl/bin/python3.11 /data2
[REDACTED] pts/17    tmux(47088).%2  10Jun26 32:14m   0.11s   0.11s -bash
[REDACTED] pts/18    tmux(47088).%3  10Jun26 16days   0.02s   0.02s -bash
[REDACTED] pts/19    tmux(47088).%4  10Jun26  4days   0.07s   0.07s -bash
[REDACTED] pts/20    tmux(47088).%5  10Jun26  7:34m   0.29s   0.29s -bash
[REDACTED] pts/6     tmux(47088).%7  Thu02   37:53m   0.04s   0.04s -bash
[REDACTED] pts/7     tmux(47088).%8  Thu02   37:52m   0.04s   0.04s -bash
```

====

```
IPv4 address for ens11f0: [REDACTED]
IPv6 address for ens11f0: fd55:faaf:e1ab:2021:8616:cff:feac:7462

=> / is using 92.2% of 294.23GB
=> /data is using 92.5% of 13.97TB
```

```
[REDACTED]-h100:~$
```

====

```
//maintenance started as scheduled with no objections raised during window of opportunity
//maintenance executing as scheduled for 6/26/2026 at 12pm US est
//users logged in with active workloads
```

```
IPv4 address for enp90s0f0np0: [REDACTED]
IPv6 address for enp90s0f0np0: fd55:faaf:e1ab:2021:ba3f:d2ff:fe4c:dbc

=> /data2 is using 90.7% of 13.86TB
```

```
[REDACTED]-b200:~$
16:16:54 up 9 days, 20:47, 3 users, load average: 6.08, 8.22, 9.29
```

USER	TTY	FROM	LOGIN@	IDLE	JCPU	PCPU	WHAT
██████████	pts/4	tmux(1389067).%1	Thu07	18:52m	0.01s	0.01s	-bash
██████████	pts/6	tmux(1389067).%0	Wed02	2days	0.00s	0.00s	-bash

@llmus03-b200:~\$

====

//maintenance started as scheduled with no objections raised during window of opportunity
 //maintenance executing as scheduled for 6/26/2026 at 12pm US est
 //users logged in with active workloads

IPv4 address for ens10f0: ██████████
 IPv6 address for ens10f0: fd55:faaf:e1ab:2021:a236:9fff:feb2:d8dc

██████████-b200:~\$ w
 16:18:07 up 2 days, 8:24, 9 users, load average: 12.64, 10.65, 9.75

USER	TTY	FROM	LOGIN@	IDLE	JCPU	PCPU	WHAT
██████████		██████████	07:26	9:42m	0.00s	0.04s	sshd: shuhq1 [priv]
██████████		██████████	07:25	9:42m	0.00s	0.04s	sshd: shuhq1 [priv]
██████████		██████████	07:16	9:42m	0.00s	0.12s	sshd: shuhq1 [priv]
██████████		██████████	05:49	9:42m	0.00s	0.11s	sshd: shuhq1 [priv]

██████████-b200:~\$

====

IPv4 address for enp90s0f0np0: ██████████
 IPv6 address for enp90s0f0np0: fd55:faaf:e1ab:2021:ac0:ebff:feff:379a

██████████-mi300x:~\$

====

//system to be reconfigured for removal of /dev/md1 (data2)
 //drives to be repurposed to system ██████████-h100

IPv4 address for ens11f0: ██████████
 IPv6 address for ens11f0: fd55:faaf:e1ab:2021:6efe:54ff:fe3d:5138

=> / is using 85.2% of 435.96GB

██████████-mi300x:~\$

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/md0	11T	2.4T	8.2T	23%	/data1

/dev/md1 11T 75G 11T 1% /data2

mi300x:~\$

mi300x:~/data1

total 12K

4.0K	drwxrwxr-x	11		4.0K	Jun 12 02:02		3
4.0K	drwxrwxr-x	16		4.0K	May 28 07:56		g
0	drwx--x---	12	root	root	217	Jun 15 15:50	docker
0	drwxrwxr-x	2		61	May 28 07:55		
0	-rwxrwxrwx	1		0	Apr 28 13:45	testfile	
4.0K	drwxrwxr-x	30		4.0K	Jun 26 16:04		

mi300x:~\$

mi300x:~/data2

total 0

mi300x:~\$

mi300x:~\$

/dev/md1:

Version : 1.2
Creation Time : Tue Apr 28 00:54:21 2026
Raid Level : raid6
Array Size : 11251445760 (10.48 TiB 11.52 TB)
Used Dev Size : 1875240960 (1788.37 GiB 1920.25 GB)
Raid Devices : 8
Total Devices : 8
Persistence : Superblock is persistent

Intent Bitmap : Internal

Update Time : Fri Jun 26 18:28:56 2026
State : clean
Active Devices : 8
Working Devices : 3
Failed Devices : 0
Spare Devices : 0

Layout : left-symmetric
Chunk Size : 512K

Consistency Policy : bitmap

Name : :1
UUID : bdf74c61:7540ae0a:7e803c78:fd0b58a0
Events : 1793

Number	Major	Minor	RaidDevice	State	
0	259	20	0	active sync	/dev/nvme8n1p1
1	259	30	1	active sync	/dev/nvme9n1p1
2	259	26	2	active sync	/dev/nvme10n1p1
3	259	34	3	active sync	/dev/nvme11n1p1
4	259	43	4	active sync	/dev/nvme12n1p1

5	259	42	5	active sync	/dev/nvme13n1p1
6	259	47	6	active sync	/dev/nvme14n1p1
7	259	44	7	active sync	/dev/nvme15n1p1

mi300x:~\$

mi300x:~\$

/dev/md0:

Version : 1.2
 Creation Time : Tue Apr 28 00:53:14 2026
 Raid Level : raid6
 Array Size : 11251445760 (10.48 TiB 11.52 TB)
 Used Dev Size : 1875240960 (1788.37 GiB 1920.25 GB)
 Raid Devices : 8
 Total Devices : 8
 Persistence : Superblock is persistent

Intent Bitmap : Internal

Update Time : Fri Jun 26 18:28:56 2026
 State : clean
 Active Devices : 8
 Working Devices : 8
 Failed Devices : 0
 Spare Devices : 0

Layout : left-symmetric
 Chunk Size : 512K

Consistency Policy : bitmap

Name : :0
 UUID : dc6a9da3:aca8d91a:247568ed:71042b22
 Events : 1814

Number	Major	Minor	RaidDevice	State	
0	259	2	0	active sync	/dev/nvme0n1p1
1	259	5	1	active sync	/dev/nvme1n1p1
2	259	11	2	active sync	/dev/nvme2n1p1
3	259	14	3	active sync	/dev/nvme3n1p1
4	259	17	4	active sync	/dev/nvme4n1p1
5	259	13	5	active sync	/dev/nvme5n1p1
6	259	23	6	active sync	/dev/nvme6n1p1
7	259	35	7	active sync	/dev/nvme7n1p1

mi300x:~\$

mi300x:~\$

/dev/md0 on /data1 type xfs

(rw,relatime,attr2,inode64,logbufs=8,logbsize=32k,sunit=1024,swidth=6144,noquota)

/dev/md1 on /data2 type xfs

(rw,relatime,attr2,inode64,logbufs=8,logbsize=32k,sunit=1024,swidth=6144,noquota)

mi300x:~\$

```

[REDACTED]-mi300x:~$
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/ubuntu-vg/lv-0 during curtin installation
/dev/disk/by-id/dm-uuid-LVM-
UIH6KHQX1T38uhtmqRbi1Gpng3ycYC0r76mX0RItYKvSkcf5wuieJtygyoePpwq8 / ext4 defaults 0
1
# /boot was on /dev/sda2 during curtin installation
/dev/disk/by-uuid/fb6bca4d-0467-4ed0-95fa-6324c0aaa653 /boot ext4 defaults 0 1
# /boot/efi was on /dev/sda1 during curtin installation
/dev/disk/by-uuid/3353-AF76 /boot/efi vfat defaults 0 1
# manual entry for the xfs vols of md0
/dev/disk/by-uuid/aa02a646-11a2-4728-b13f-43daea9be1f3 /data1 xfs
defaults,nofail 0 2
# manual entry for the xfs vols of md1
/dev/disk/by-uuid/6f1ee2f2-d6cc-48d1-beb1-7859735f78a0 /data2 xfs
defaults,nofail 0 2
/swap.img none swap sw 0 0
[REDACTED]-mi300x:~$

```

```

[REDACTED]-mi300x:~$
Personalities : [raid6] [raid5] [raid4] [linear] [multipath] [raid0] [raid1]
[raid10]
md1 : active raid6 nvme13n1p1[5] nvme12n1p1[4] nvme8n1p1[0] nvme14n1p1[6]
nvme15n1p1[7] nvme10n1p1[2] nvme11n1p1[3] nvme9n1p1[1]
11251445760 blocks super 1.2 level 6, 512k chunk, algorithm 2 [8/8]
[UUUUUUUU]
bitmap: 0/14 pages [0KB], 65536KB chunk

md0 : active raid6 nvme2n1p1[2] nvme5n1p1[5] nvme4n1p1[4] nvme3n1p1[3] nvme0n1p1[0]
nvme6n1p1[6] nvme1n1p1[1] nvme7n1p1[7]
11251445760 blocks super 1.2 level 6, 512k chunk, algorithm 2 [8/8]
[UUUUUUUU]
bitmap: 0/14 pages [0KB], 65536KB chunk

unused devices: <none>
[REDACTED]-mi300x:~$

```

```

[REDACTED]-mi300x:~$
# mdadm.conf
#
# !NB! Run update-initramfs -u after updating this file.
# !NB! This will ensure that initramfs has an uptodate copy.
#
# Please refer to mdadm.conf(5) for information about this file.
#

# by default (built-in), scan all partitions (/proc/partitions) and all
# containers for MD superblocks. alternatively, specify devices to scan, using

```

```
# wildcards if desired.
#DEVICE partitions containers

# automatically tag new arrays as belonging to the local system
HOMEHOST <system>

# instruct the monitoring daemon where to send mail alerts
MAILADDR root

# definitions of existing MD arrays

# This configuration was auto-generated on Wed, 11 Sep 2024 14:22:21 +0000 by
mkconf
ARRAY /dev/md0 metadata=1.2 name=llmus08:0 UUID=dc6a9da3:aca8d91a:247568ed:71042b22
ARRAY /dev/md1 metadata=1.2 name=llmus08:1 UUID=bdf74c61:7540ae0a:7e803c78:fd0b58a0
[REDACTED]-mi300x:~$
```

```
[REDACTED]-mi300x:~$
umount: /data2 unmounted
@llmus08-mi300x:~$
Filesystem                                Size  Used Avail Use% Mounted on
/dev/md0                                  11T   2.4T   8.2T   23% /data1
[REDACTED]-mi300x:~$
```

```
[REDACTED]-mi300x:~$
/dev/md0 on /data1 type xfs
(rw,relatime,attr2,inode64,logbufs=8,logbsize=32k,sunit=1024,swidth=6144,noquota)
[REDACTED]-mi300x:~$
```

```
[REDACTED]-mi300x:~$
mdadm: stopped /dev/md0
[REDACTED]-mi300x:~$
```

```
[REDACTED]-mi300x:~$
Personalities : [raid6] [raid5] [raid4] [linear] [multipath] [raid0] [raid1]
[raid10]
md0 : active raid6 nvme2n1p1[2] nvme5n1p1[5] nvme4n1p1[4] nvme3n1p1[3] nvme0n1p1[0]
nvme6n1p1[6] nvme1n1p1[1] nvme7n1p1[7]
11251445760 blocks super 1.2 level 6, 512k chunk, algorithm 2 [8/8]
[UUUUUUUU]
    bitmap: 0/14 pages [0KB], 65536KB chunk

unused devices: <none>
[REDACTED]-mi300x:~$
```

```
=====
```


====

//note the ip change for the following systems

██████-h100 > ██████████

====

systems verified back up:

\$ ssh @██████████

IPv4 address for ens11f0np0: ██████████

IPv6 address for ens11f0np0: fd55:faaf:e1ab:2021:ba3f:d2ff:fe90:addc

=> / is using 90.8% of 195.80GB

Last login: Fri Jun 26 20:52:56 2026

██████-h200:~\$

\$ ssh ██████████

IPv4 address for ens11f0: ██████████

IPv6 address for ens11f0: fd55:faaf:e1ab:2021:8616:cff:feac:7462

=> / is using 92.0% of 294.23GB

=> /data is using 92.5% of 13.97TB

Last login: Fri Jun 26 20:55:20 2026

██████-h100:~\$

\$ ssh @██████████

IPv4 address for enp90s0f0np0: ██████████

IPv6 address for enp90s0f0np0: fd55:faaf:e1ab:2021:ba3f:d2ff:fe4c:dbc

=> /data2 is using 91.3% of 13.86TB

Last login: Fri Jun 26 20:58:36 2026

██████-b200:~\$

\$ ssh @██████████

IPv4 address for ens10f0: ██████████

IPv6 address for ens10f0: fd55:faaf:e1ab:2021:a236:9fff:feb2:d8dc

Last login: Fri Jun 26 19:50:30 2026 from ██████████

██████████-b200:~\$

\$ ssh @██████████

IPv4 address for enp90s0f0np0: ██████████

IPv6 address for enp90s0f0np0: fd55:faaf:e1ab:2021:ac0:ebff:feff:379a

Last login: Fri Jun 26 21:22:17 2026 from ██████████

██████████-mi300x:~\$

\$ ssh @██████████

IPv4 address for ens11f0: ██████████

IPv6 address for ens11f0: fd55:faaf:e1ab:2021:6efe:54ff:fe3:5138

=> / is using 85.2% of 435.96GB

Last login: Fri Jun 26 18:27:59 2026 from ██████████

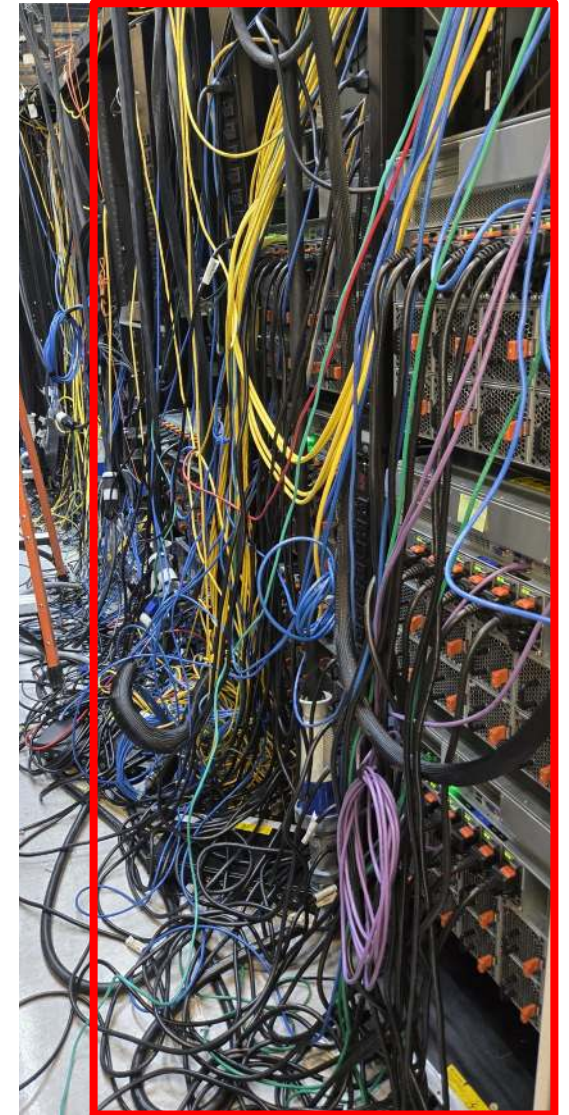
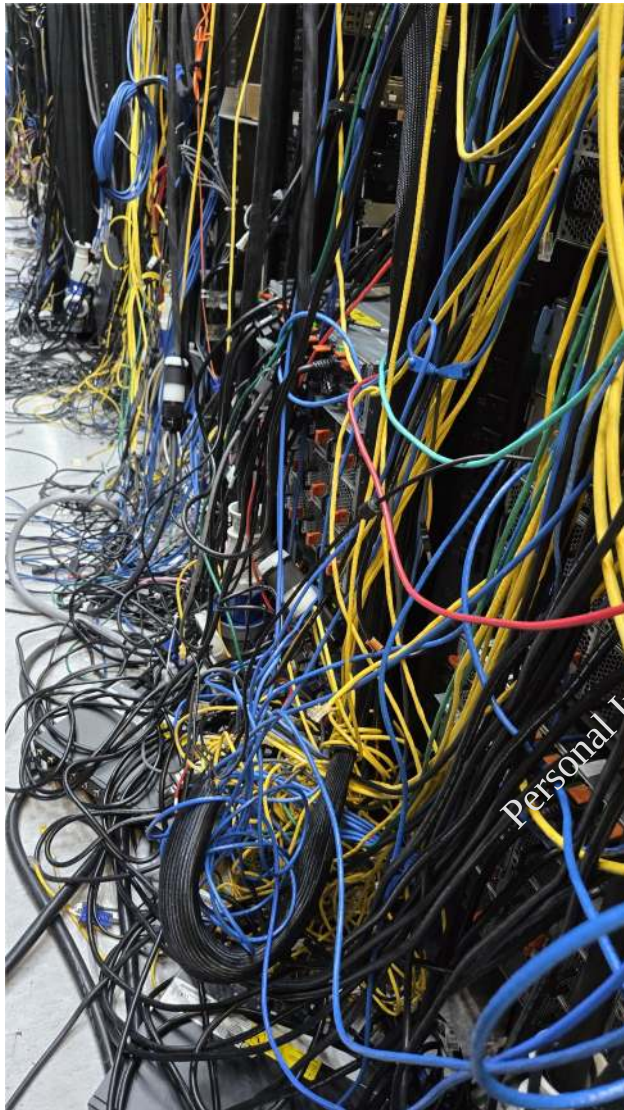
██████████-mi300x:~\$

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My inheritance:

What I had to clean up and redeploy or remedy as best effort given the constraints and authorization

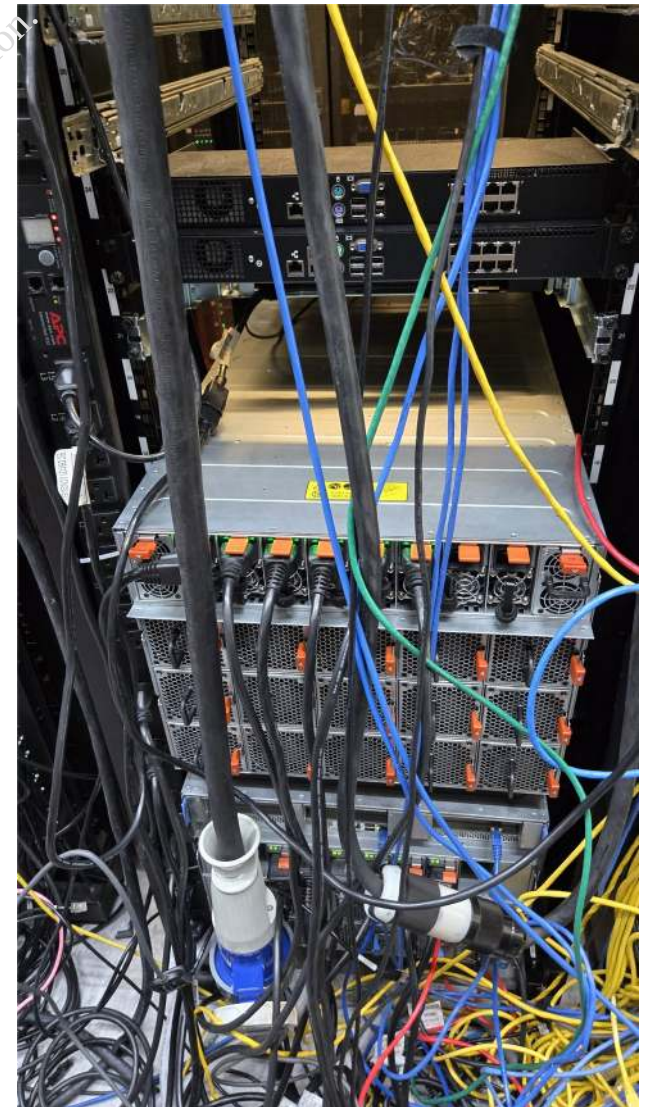
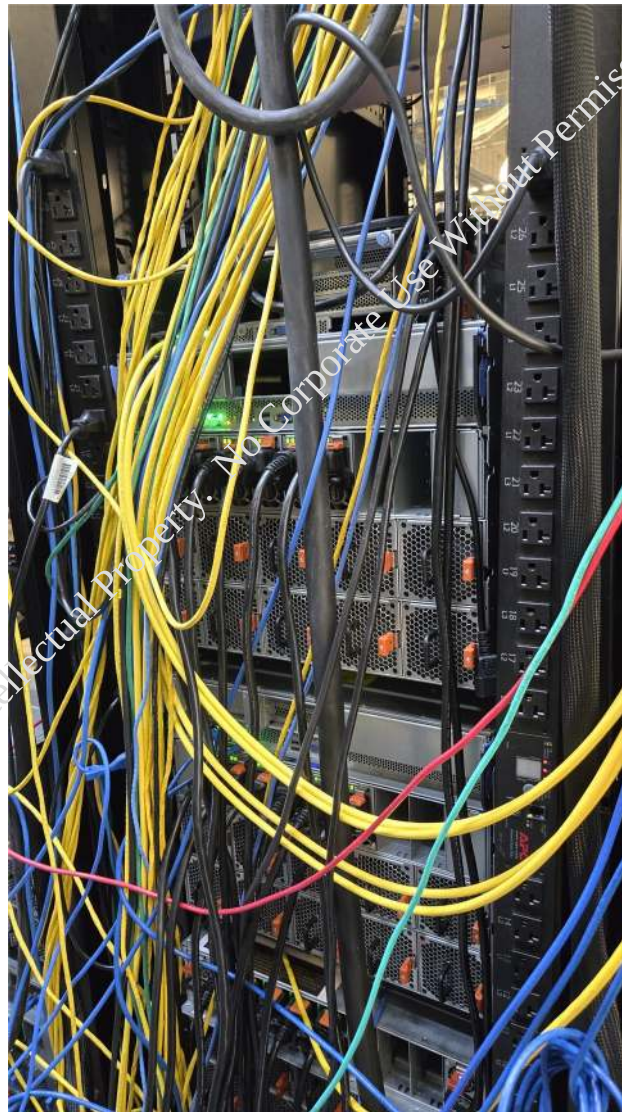
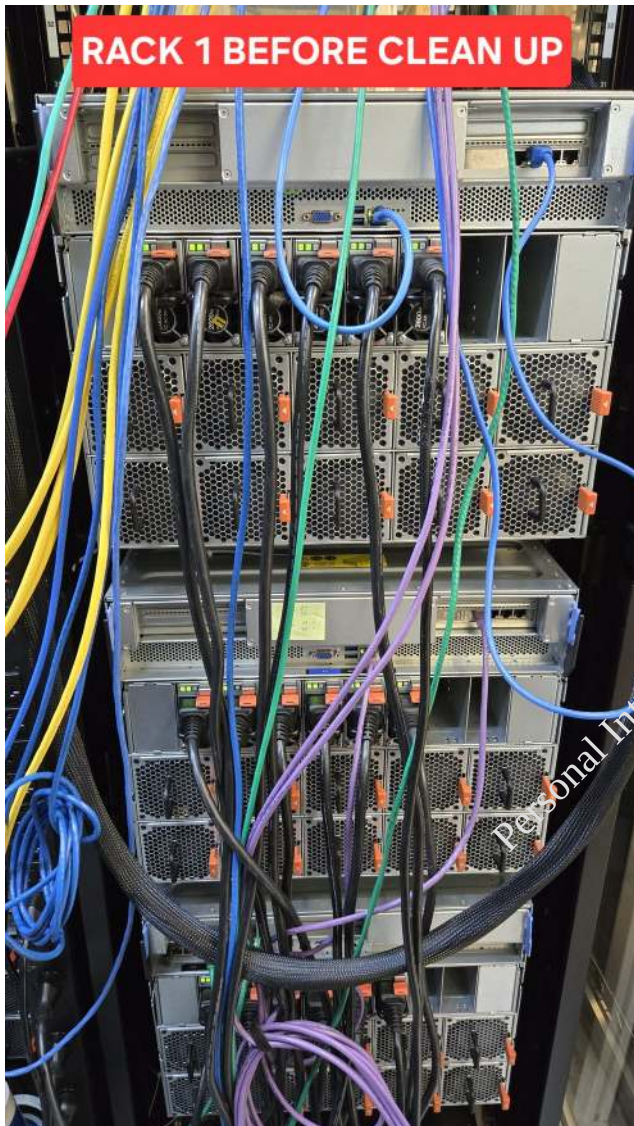
Initial fleet of 10, expanding into 13~16, each 8 gpu cluster system ~\$1M cost/value



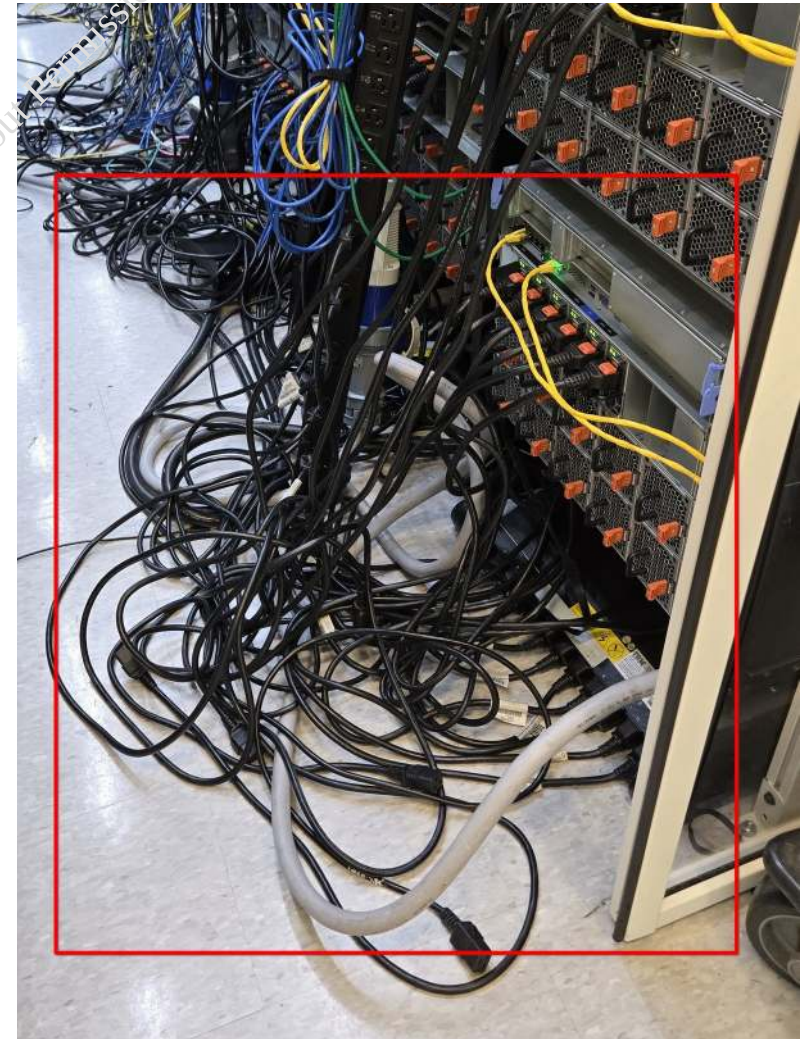
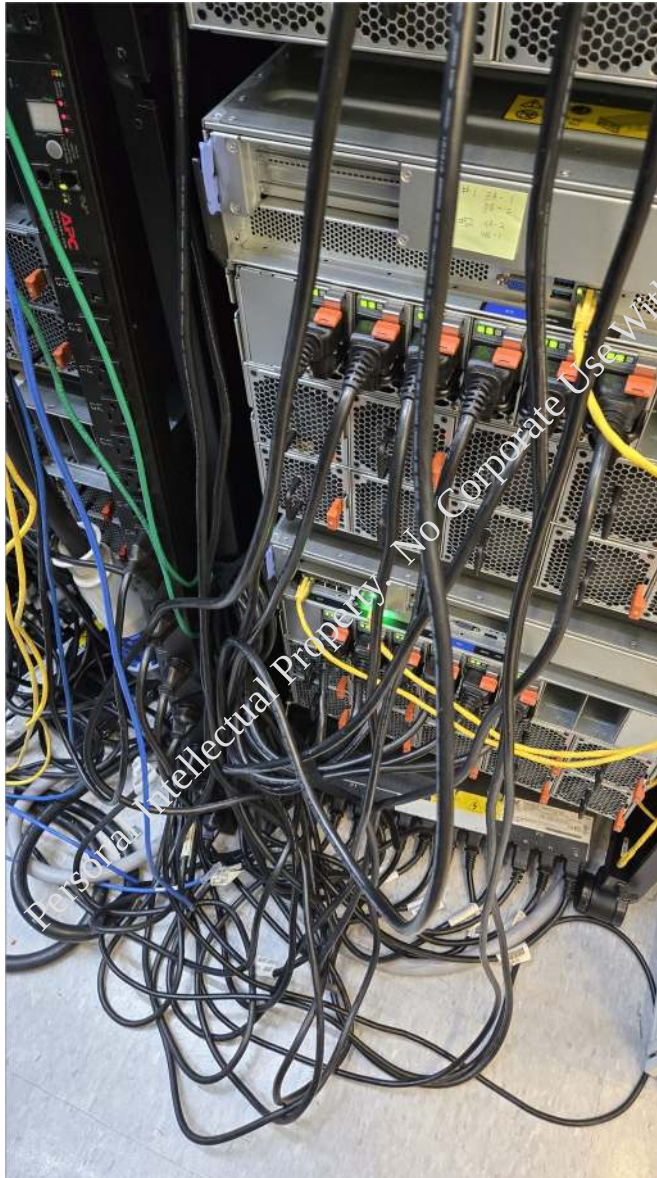
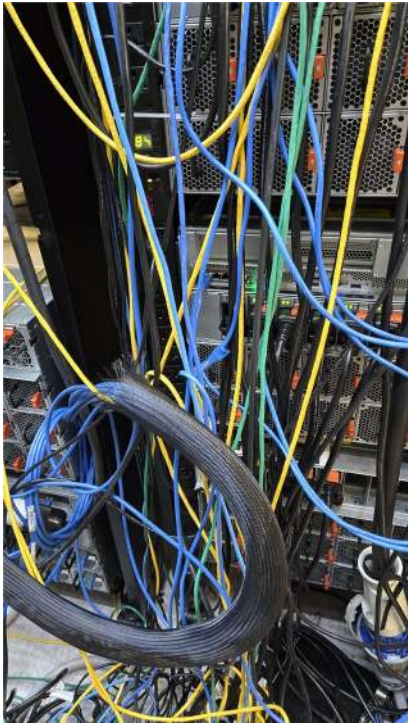
Thermal issues from the back and front, hot zones, no clearance for the engineered airflow

The spaghetti cabling, nothing traceable, no labels, no standard best practices of cabling lengths, added troubleshooting due to unable to rule out any known good gear (cabling)

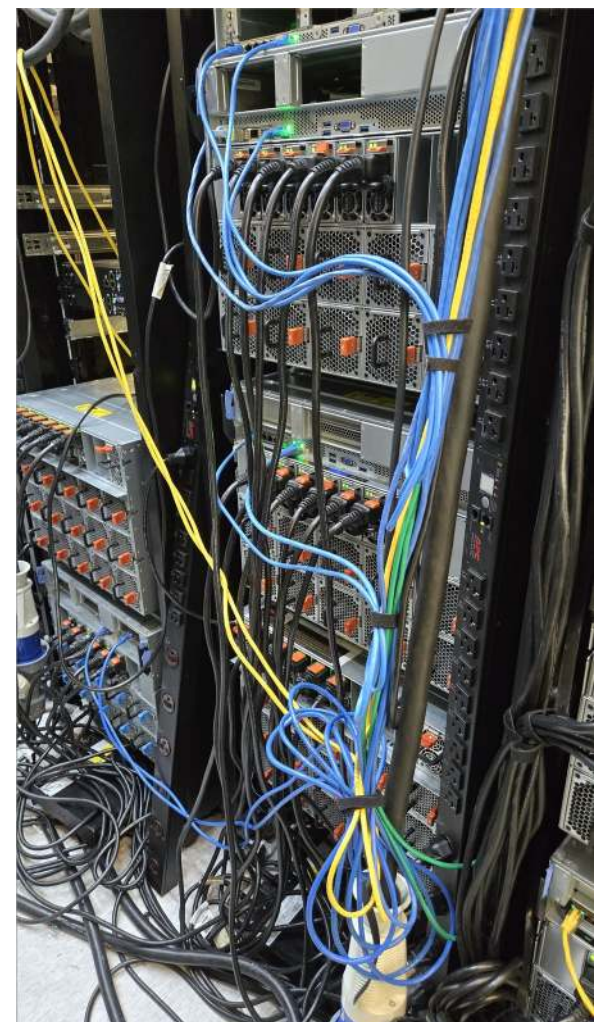
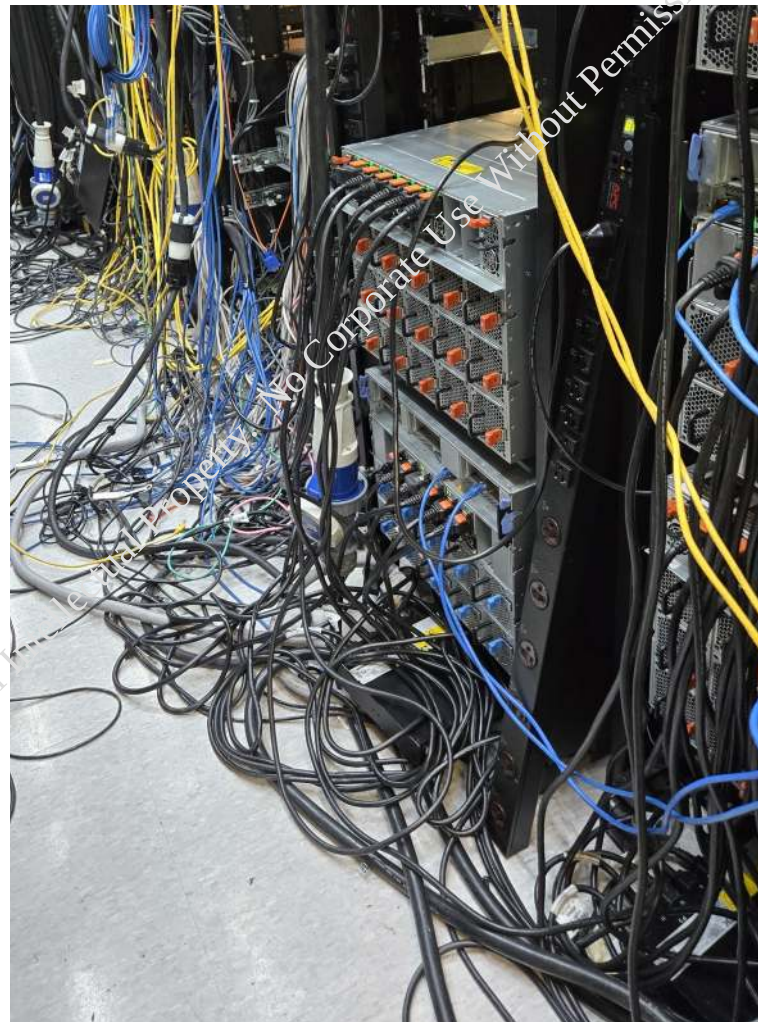
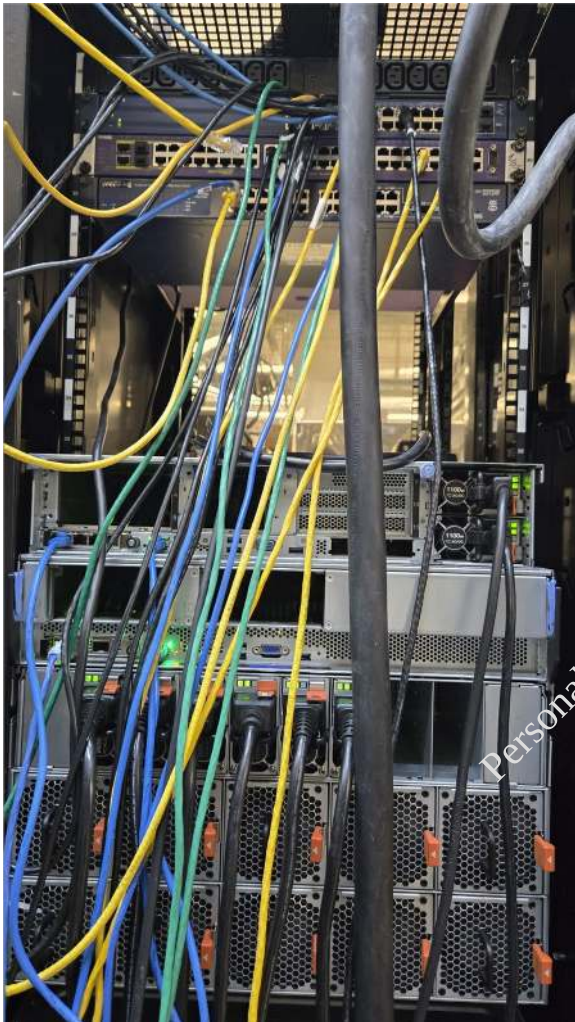
The first 3 racks before any clean up attempts...



The trip hazards, blockage of limited aisle space, along with the hot spot zones, increased fan utilization of hot air, melting the cables blocking the airflow



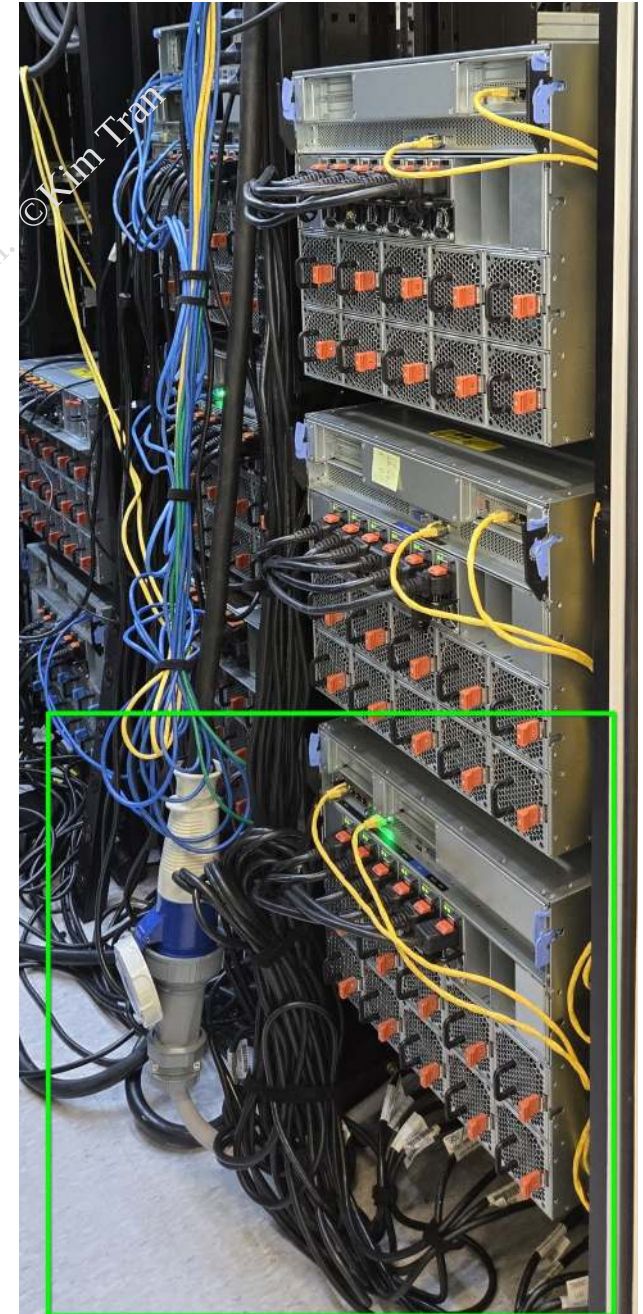
The system chassis / equipment is not in the proper rack size to accommodate the depth making any proper best practices cable management / air flow a best effort attempt given the constraints of the environment



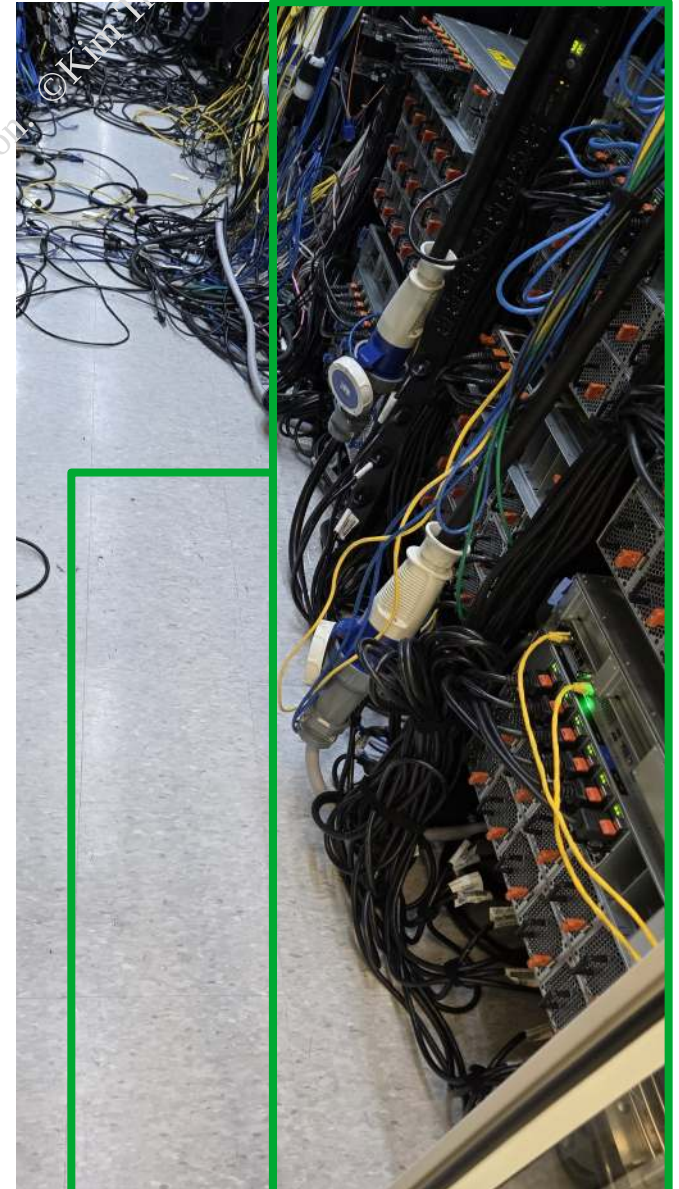
Working on remedying the network cables for some consistency on cable length, and switch port structure and population scheme



Redeployed physical cabling, decoupled inter-system connection dependencies, remedy the thermal air flow issues, given psu limitations, reclaimed limited aisle space



Redeployed cabling for racks 2 and 3, remedy the thermal air flow issues and reclaimed what little aisle space available



Remedy the internal air flow / pressure as designed and engineered

